



SRED - Thursday 701

GROUP ASSIGNMENT(2026-01)

Academic Honesty Policy Statement

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1.0 Executive Summary

This report confirms that building an AI-Assisted Diagnostic tool is both necessary and possible (feasible). Our study reveals that our current traffic systems use the fixed timer that cannot adapt to real-time conditions. This causes significant problems, wasting both time and money. Moreover, it causes significant problems during an emergency situation or event such as ambulances and fire trucks need to pass quickly. In addition, it also adversely affects the environment making the city look messy and overcrowded, and harms the earth as well due to excessive fuel consumption and vehicle emissions.

Key Findings:

- **Critical Requirements:** The system needs to respond in real-time and super-fast (NFR-1: less than 500ms latency) and stay up almost all the time (NFR-2: over 99.95% reliable) are the most demanding requirements.
- **Architecture:** Use a flexible, real-time microservices design with on-site computing at each intersection. It handles local fixes (like adjusting signals on each spot) and uses the cloud to smartly manage traffic across the whole city.
- **Top Risk:** Hooking it up to old traffic systems could be difficult and could cause issues. We'll test it early with small trials and set aside money to replace outdated gear.
- **Recommendation:** Start right away with 10 test intersections to prove it works well, then move it out to the hold city over 18-24 months, focusing on the busiest spots.

2.0 Phase 1: Domain Mastery and Problem Definition

2.1 Domain Analysis and Context Setting

Problem Description

Modern cities face a critical transportation crisis. Urban traffic congestion costs major metropolitan areas billions of dollars annually in lost productivity, wasted fuel, and environmental damage. In a typical city of 500,000 residents, commuters waste an average of 50-70 hours per year sitting in traffic, translating to approximately \$150-200 million in economic losses.

Current traffic management systems, many of which were installed 20-30 years ago, operate on fixed timing patterns programmed during installation. These systems give the same green light duration at 8:00 AM whether there are 2 vehicles or 200 vehicles waiting. They cannot respond to accidents, special events, weather conditions, or unexpected congestion in real-time.

Critical Gaps in Current Systems:

- **No Real-Time Adaptability:** Cannot adjust to changing conditions throughout the day
- **Lack of Network Coordination:** Each intersection optimizes locally, creating bottlenecks elsewhere
- **No Predictive Capabilities:** Cannot forecast problems before they occur
- **Poor Information for Citizens:** Drivers navigate without accurate real-time data
- **Limited Emergency Support:** No priority routing for ambulances and fire trucks
- **No Data for Planning:** Transportation planners lack comprehensive data for infrastructure investment decisions

The result is a transportation network operating at only 60-70% of its theoretical capacity, with cascading impacts on emergency response times, air quality, and quality of life. **Existing**

Solutions and Industry Standards

Current Systems in Use: Several cities have deployed partial smart traffic solutions, though most remain limited in scope and capability:

SCOOT (Split Cycle Offset Optimization Technique): Deployed in London, Dublin, and other European cities. Adjusts signal timings every few minutes based on real-time traffic measurements. However, it requires expensive proprietary hardware (\$30,000-50,000 per intersection) and lacks modern predictive analytics and AI capabilities. The system is over 30 years old with limited integration to mobile apps or navigation services.

SCATS (Sydney Coordinated Adaptive Traffic System): Used in Sydney, Australia and parts of the United States. Provides adaptive control that responds to traffic demand patterns. Limitations include reliance on legacy hardware, minimal cloud integration, and high maintenance costs. The system cannot predict future congestion or integrate with modern IoT sensor networks.

InSync by Rhythm Engineering: More modern adaptive system using high-resolution radar sensors for real-time vehicle detection. Offers excellent local optimization with vehicle by tracking. However, the cost is prohibitive at \$50,000-80,000 per intersection, preventing city-wide deployment in most municipalities. Limited network-wide coordination capabilities.

Common Limitations:

- High per-intersection costs (\$30,000-80,000) prevent comprehensive city-wide deployment
- Proprietary vendor-locked systems with expensive ongoing licensing fees
- Limited or no AI/machine learning for traffic prediction and pattern recognition
- Poor integration with citizen-facing mobile applications
- Weak data analytics and long-term planning capabilities
- Difficulty integrating with emergency dispatch and public transit systems

Regulatory Environment

Traffic Standards:

- NTCIP: Industry standard for traffic device communication
- IEEE 1512: Incident management protocols
- MUTCD: Federal traffic signal standards

Privacy Requirements:

- GDPR/CCPA compliance for data protection Anonymous data collection
- (no personal tracking) Vehicle data anonymized within 60 seconds

Accessibility:

- ADA compliance for mobile applications WCAG 2.1 Level AA standards

2.2 Problem Statement and Scope

In rapidly urbanizing environments, escalating traffic congestion poses a multifaceted challenge, leading to economic losses, environmental degradation, public health risks, and diminished quality of life. This project addresses the core issue by developing a secure, cloudbased Smart City Traffic Management System (SCTMS) that leverages real-time sensor data, advanced artificial intelligence (AI), and machine learning algorithms to optimize traffic flows dynamically. The primary objectives are to reduce average travel times by 15-25% during peak hours, enhance emergency response times by 20-30% through prioritized routing, and minimize overall urban mobility inefficiencies. By integrating IoT-enabled infrastructure with predictive modeling, the system will process petabytes of data daily from diverse sources, ensuring scalability, cybersecurity compliance (e.g., NIST frameworks), and seamless interoperability with existing municipal systems. This initiative aligns with global sustainability goals, such as UN Sustainable Development Goal 11 (Sustainable Cities and Communities), by fostering resilient, equitable, and efficient transportation networks.

Project Justification

Urban traffic congestion is not merely an inconvenience but a systemic crisis that drains resources and hampers progress. In a typical mid-sized city like ours, commuters lose over 50 hours annually to gridlock, contributing to heightened stress levels, increased fuel consumption, and elevated greenhouse gas emissions. The SCTMS represents a proactive, technology-driven intervention to reclaim these lost efficiencies. By harnessing AI for real

time decision-making, the system will transform reactive traffic management into a predictive, adaptive ecosystem, yielding measurable socioeconomic and environmental dividends.

Key Benefits:

- **Economic Savings:** Economic Savings: Projected annual reduction in congestion-related costs by \$30-40 million, encompassing fuel savings, lost productivity, and vehicle maintenance expenses. This is based on conservative estimates from similar deployments in cities like Singapore and Barcelona, where AI-optimized systems have recouped investments within 2-3 years.
- **Travel Time Reduction:** Achieve 15-25% decrease in average commute durations through adaptive signal timing and route optimization, benefiting over 1 million daily users and improving workforce productivity.
- **Enhanced Emergency Response:** 20-30% faster arrival times for critical services (e.g., ambulances reaching scenes in under 5 minutes on average), potentially saving hundreds of lives annually by minimizing delays in high-traffic zones.
- **Environmental Impact:** 10-15% reduction in vehicle emissions (CO₂, NO_x, and particulate matter) via smoother traffic flows and eco-routing, supporting compliance with air quality standards like those set by the EPA or EU directives.
- **Long-Term ROI:** Over a 10-year horizon, deliver a 3-5x return on the \$80-120 million initial investment, factoring in operational efficiencies, reduced infrastructure wear, and potential revenue from data licensing to urban planners or third-party apps. This ROI model incorporates sensitivity analyses for variables such as traffic growth rates (assumed at 2-3% annually) and technology adoption curves. These benefits are substantiated by pilot studies from the U.S. Department of Transportation's Intelligent Transportation Systems program, which reported similar gains in cities like Los Angeles and Seattle after implementing comparable AI-driven platforms.
- **Scope Definition: In-Scope vs. Out-of-Scope**

To ensure project success, clarity on deliverables is paramount. The following delineates the boundaries, focusing resources on high-impact features while explicitly excluding peripheral functionalities to maintain feasibility, budget adherence, and timeline efficiency (target completion: 24 months for Phase 1 rollout).

In-Scope What We Promise to Delivery

The core deliverables encompass a comprehensive, integrated platform emphasizing data-driven optimization and user-centric tools. All components will be developed using agile methodologies, with cloud infrastructure (e.g., AWS or Azure) ensuring 99.99% uptime and HIPAA/GDPR-compliant data handling.

Real-Time Traffic Monitoring: Deployment of sensors (e.g., inductive loops, radar, and LiDAR) and high-resolution cameras at 500+ intersections to capture vehicle counts, speeds, occupancy, and pedestrian volumes. Data ingestion via edge computing will enable sub-second latency processing.

- **Adaptive Signal Control:** AI algorithms (e.g., reinforcement learning models) to dynamically adjust signal phases, cycle lengths, and offsets based on real-time conditions, incorporating multi-modal data like weather feeds and event calendars for proactive adjustments.
- **Incident Detection:** Computer vision and anomaly detection ML models to identify accidents, breakdowns, or hazards within 60 seconds of occurrence, triggering automated alerts to operators and integrating with 911 dispatch systems.
- **Emergency Vehicle Preemption:** GPS-enabled priority routing for authorized vehicles (police, fire, ambulances), using vehicle-to-infrastructure V2 communication to clear paths via phased green lights and dynamic lane assignments, with audio-visual cues for Drivers.
- **Predictive Analytics:** Time-series forecasting using LSTM neural networks to predict traffic volumes and bottlenecks 30, 60, and 120 minutes ahead, with accuracy targets of 85-90%, enabling preemptive interventions like temporary lane reversals.
- **Citizen Mobile App:** A user-friendly iOS/Android application providing real-time traffic visualizations, personalized route suggestions via graph-based algorithms (e.g., A* with traffic weights), incident alerts via push notifications, and gamified incentives for offpeak travel.
- **Public Transit Integration:** Transit signal priority (TSP) for buses delayed by >2 minutes, using AVL (automatic vehicle location) data to extend green phases, improving

On time performance by 15-20% and encouraging modal shifts from private vehicles.

- **Traffic Operator Dashboard:** A web-based command center with GIS mapping, live video feeds, customizable KPIs (e.g., delay indices), and manual override capabilities, supporting multi-user access with role-based permissions for up to 50 operators.
- **Data Analytics:** Robust backend for historical trend analysis using tools like Apache Spark, generating reports on peak patterns, equity metrics (e.g., access disparities by neighborhood), and scenario simulations to inform long-term infrastructure investments.
- **Environmental Monitoring:** Integration of air quality sensors (e.g., for PM2.5, O3 at key nodes, correlating pollution spikes with traffic density via regression models to support green policy decisions and public health advisories.

Geographic Coverage: Phase 1 rollout targets 500 intersections in the downtown core (approximately 50 square kilometers), prioritizing high-density arterials and commercial districts. This includes scalability provisions for expansion to 2,000 intersections city-wide by Phase 3 (Year 5), with modular hardware kits for rapid deployment and API endpoints for future zoning extensions.

Out-of-Scope What We Will NOT Deliver:

To avoid scope creep and focus on transportation optimization, the following elements are explicitly excluded. These may be addressed through future partnerships or separate initiatives, but they fall outside the SCTMS's primary remit. Additional considerations include regulatory, ethical, and operational boundaries to prevent overreach into sensitive areas like privacy, finance, or law enforcement, ensuring the project remains laser-focused on core traffic efficiency goals.

Autonomous Vehicle Control: The system will disseminate traffic data via standardized APIs (e.g., V2X protocols) but will not interface with or control individual autonomous vehicles, deferring to OEM-specific software for safety-critical maneuvers. This exclusion mitigates liability risks associated with vehicle autonomy standards (e.g., SAE Levels 4-5) and avoids dependency on evolving AV ecosystems.

- **Traffic Law Enforcement:** No deployment of red-light or speed cameras for automated ticketing; enforcement remains a municipal police function, with the SCTMS

limited to providing evidentiary data feeds upon request. This keeps the system neutral and avoids contentious debates over surveillance and civil liberties.

- **Toll Collection:** Exclusion of any electronic tolling infrastructure, payment gateways, or dynamic congestion pricing mechanisms, which require separate regulatory approvals and financial auditing. Future integrations could explore APIs for third-party toll operators, but these are not part of the baseline delivery
 - **Vehicle Registration/DMV Functions:** No integration with departmental motor vehicle registries for licensing, emissions testing, or ownership verification; user data will remain anonymized to privacy standards. This boundary protects against data silos and complies with data minimization principles under frameworks like CCPA.
 - **Private Parking Management:** The platform will not handle smart parking reservations, garage access controls, or valet integrations, focusing instead on street-level flow; off-street solutions can leverage exported aggregate data. Partnerships with parking apps (e.g., via open data portals) may be recommended post-launch for holistic mobility.
 - **Personal Vehicle Tracking:** Strict adherence to anonymization protocols (e.g., differential privacy techniques) ensures only aggregate, non-identifiable data is processed—no individual tracking, geofencing, or behavioral profiling of private vehicles. This upholds ethical AI practices and avoids GDPR-like fines from granular user surveillance.
 - **Road Construction Management:** While the SCTMS will incorporate construction zone data from external feeds (e.g., 511 APIs) for detour routing, it will not encompass full project lifecycle management, permitting workflows, or contractor coordination—these remain under public works jurisdiction. Enhanced features could include predictive impact modeling for construction schedules in future phases, but only as advisory tools.
- This scoped approach mitigates risks such as integration complexities and regulatory hurdles, while laying a foundation for iterative enhancements based on post-deployment feedback and performance metrics. Regular scope reviews will be conducted quarterly to

adapt to emerging technologies like 6G connectivity or quantum-secure encryption without compromising the core deliverables.

2.3 Simulated Elicitation Plan

Stakeholder Groups and Methods

We identified 7 key stakeholder groups requiring different elicitation approaches:

Stakeholder 1: City Traffic Engineers & Operators

Role: Primary system users managing day-to-day traffic operations.

Elicitation Methods:

- Job Shadowing (3 days observing control center)
- Structured Interviews (90 minutes with 5-7 operators)
- Workflow Mapping

Sample Questions:

1. Walk me through a typical shift. What are your most time-consuming tasks?
2. When an incident occurs, what information do you need immediately?
3. What decisions could be automated versus requiring human judgment?
4. What's the most frustrating limitation of the current system?
5. How do you measure success in traffic management?

Stakeholder 2: Emergency Services (Police, Fire, EMS)

Role: Require priority routing and signal preemption for emergency responses.

Elicitation Methods:

- Focus Groups (joint sessions with all emergency departments)
- Ride-Alongs (6 shifts with different services)
- Scenario Analysis

Sample Questions:

1. How much time do you lose to traffic during emergencies?
2. What route information would help you reach incidents faster?
3. How should the system prioritize multiple simultaneous emergency requests?

4. What interface works best for field personnel under stress?
5. What happens if the system fails during a major incident?

Stakeholder 3: Public Transit Authority

Role: Seeking improved bus on-time performance through signal priority.

Elicitation Methods: Requirements Workshop (full-day collaborative session)

System Integration Review Passenger Journey Mapping Sample Questions: 1. What percentage of delays are caused by traffic congestion? 2. Which routes have the biggest reliability problems? 3. How should the system balance transit priority with general traffic flow? 4. What real-time data do you share with passengers? 5. What performance metrics would demonstrate value?

Stakeholder 4: Citizens & Commuters

Role: End users of roads and consumers of traffic information.

Elicitation Methods:

Online Survey (5,000+ residents) User Persona Development (5-7 representative profiles)

Mobile App Prototype Testing Community Town Halls

Sample Questions:

1. How do you currently get traffic information?
2. What would make you change your route or departure time?
3. How accurate do travel time estimates need to be for you to trust them?
4. What concerns do you have about the city collecting traffic data?
5. Would you share your location for better traffic information?

Stakeholder 5: City Leadership & Budget Authority

Role: Decision-makers approving funding and accountable to taxpayers. Elicitation Methods:

Executive Briefings (monthly updates) Comparative Analysis (case studies from other cities)

Financial Modeling (10-year TCO and ROI)

Sample Questions:

1. What are the city's top 3 transportation priorities?
2. How will you measure project success?
3. What's the acceptable timeline for measurable results?

4. What political concerns must be addressed proactively?
5. What reporting will you need for the city council?

Stakeholder 6: IT/Infrastructure Department

Role: Responsible for deployment, security, and system maintenance.

Elicitation Methods:

Technical Architecture Review Site Surveys (data centers, network capacity) Vendor Demonstrations

Sample Questions:

1. What are the network infrastructure constraints?
2. What security standards must we meet?
3. Describe your disaster recovery capabilities.
4. What's your experience with IoT device management?
5. What concerns do you have about vendor lock-in?

Stakeholder 7: Environmental & Urban Planning Agencies

Role: Focus on sustainability, air quality, and long-term planning.

Elicitation Methods: Data Requirements Workshop Policy Review (climate plans, transportation plans)

Sample Questions:

1. What environmental metrics must you track?
2. How would traffic data improve air quality monitoring?
3. What data would help long-term infrastructure planning?
4. How should the system support alternative transportation (bikes, pedestrians)?
5. What equity concerns exist in underserved neighborhoods?

References Link:

1. **Revolutionizing Urban Mobility with AI, IoT, and Predictive Analytics** – A *systematic review* showing how AI and IoT improve traffic flow, adaptive control, and prediction — directly supporting your claims about AI-driven optimization and congestion reduction. <https://www.mdpi.com/2079-9292/14/4/719?utm>

2. **Intelligent Network Traffic Management in Smart Cities Using AI** – Academic article on integrating AI and IoT for adaptive, real-time traffic control, perfectly aligned with your SCTMS description. <https://ijctece.com/index.php/IJCTEC/article/view/58?utm>
3. **Design & Implementation of an ML and IoT Based Adaptive Traffic Management System** – Practical research on adaptive ML-powered signal control and congestion reduction, matching your real-time monitoring and control modules. <https://www.mdpi.com/1424-8220/22/8/2908?utm>
4. **An Optimal Control Strategy for Emergency Vehicle Priority Systems** – A *ScienceDirect* article focused on emergency vehicle preemption using IoT and edge computing, supporting your emergency response routing objectives. <https://www.sciencedirect.com/science/article/pii/S2665917423000338?utm>

3.0 Phase 2: Requirements Analysis and Modeling

This phase focuses on identifying, analyzing, and formalizing the functional requirements of the **Smart City Traffic Management System (SCTMS)**. The goal is to ensure that the system meets stakeholder needs, supports efficient traffic operations, and provides scalable, reliable services for real-time and long-term traffic management. The requirements defined in this section form the foundation for system design, implementation, and validation.

3.1 Core Requirements Package

3.1.1 Functional Requirements (FRs)

FR-1: Real-Time Sensor Data Collection (Priority: HIGH)

Traffic data shall be continuously collected from intersection sensors with a minimum frequency of 1 second for vehicle detection and 5 seconds for aggregated metrics, including

vehicle count, average speed, and lane occupancy. This real-time data supports adaptive signal control, incident detection, and accurate traffic visualization. Multiple sensor types, such as inductive loop detectors, cameras, radar, and IoT-based traffic sensors, must be supported.

FR-2: Adaptive Traffic Signal Control (Priority: HIGH)

The system shall automatically calculate and adjust traffic signal timings based on real-time traffic conditions. Using traffic flow data, queue lengths, and congestion levels, the system will optimize signal phases to improve network-wide throughput, reduce delays, and minimize stop-and-go traffic. Adaptive control should function continuously without manual intervention while allowing fallback to predefined signal plans when required.

FR-3: Automatic Incident Detection (Priority: HIGH)

The system detects traffic incidents such as accidents, stalled vehicles, or abnormal congestion patterns within 60 seconds. This detection will be achieved using anomaly detection techniques applied to traffic flow, speed variance, and occupancy data. Rapid incident detection enables quicker response times, reduces secondary accidents, and improves overall road safety.

FR-4: Emergency Vehicle Preemption (Priority: HIGH)

Real-time location data from emergency vehicles (e.g., ambulances, fire trucks, and police vehicles) shall be received and processed to automatically adjust traffic signals along designated routes. Signal preemption must be executed within a maximum response time of 3 seconds to minimize delays for emergency services while maintaining traffic safety.

FR-5: Predictive Traffic Forecasting (Priority: MEDIUM)

Short-term traffic forecasts of 30 minutes, 60 minutes, and 120 minutes shall be generated using machine learning models trained on historical traffic data combined with real-time inputs. These forecasts support proactive traffic management decisions and provide predictive travel information to road users.

FR-6: Dynamic Route Optimization (Priority: MEDIUM)

Optimal travel routes shall be calculated and recommended during congestion events. Route optimization algorithms will incorporate real-time traffic conditions, detected incidents, and historical travel patterns, with the goal of achieving at least a 10% reduction in travel time compared to standard routing methods.

FR-7: Citizen Mobile Application (Priority: HIGH)

Mobile applications for IOS and Android platforms shall be made available to citizens. These applications will present real-time traffic maps, estimated travel times, congestion levels, and incident alerts. Displayed data must be refreshed at intervals of no more than 30 seconds to ensure accuracy and timeliness.

FR-8: Incident Alert Notifications (Priority: MEDIUM)

Push notifications shall be delivered to users within 2 minutes of detecting incidents that impact their saved or frequently used routes. Alerts should provide relevant details, including incident type, location, expected delay, and suggested alternative routes when available.

FR-9: Traffic Management Center Dashboard (Priority: HIGH)

A secure, web-based dashboard shall be provided for traffic management center operators. The dashboard will display real-time network status, live camera feeds, detected incidents, congestion heatmaps, and system alerts, along with access to analytics tools and operational controls.

FR-10: Manual Signal Override (Priority: HIGH)

Authorized operators shall be able to manually override traffic signal operations during special events, emergencies, or system failures. Manual overrides must take effect within 10 seconds of operator input and be logged for auditing and performance evaluation purposes.

FR-11: Public Transit Signal Priority (Priority: MEDIUM)

Signal priority shall be granted to public transit vehicles, particularly buses running more than 3 minutes behind schedule. This capability is intended to improve public transport reliability and promote the use of sustainable transportation options.

FR-12: Environmental Data Integration (Priority: LOW)

Environmental data including air quality and noise levels, shall be collected through integrated sensors. This information will be correlated with traffic conditions to support environmental impact analysis and smart city sustainability initiatives.

FR-13: Historical Data Storage (Priority: MEDIUM)

All collected traffic historics and environmental data shall be stored for a minimum of 5 years in a structured and queryable format. Long-term data retention enables historical analysis, machine learning model training, policy assessment, and regulatory reporting.

FR-14: Automated Performance Reporting (Priority: MEDIUM)

Automated reports shall be generated on a daily, weekly, and monthly basis, summarizing key performance indicators (KPIs) such as average travel time, congestion levels, incident response time, and signal efficiency.

FR-15: Third-Party Navigation API (Priority: MEDIUM)

Public APIs shall be provided to allow third-party navigation and mobility applications to access real-time and historical traffic data. These APIs must enforce authentication, authorization, and rate-limiting mechanisms to ensure system security and fair usage.

Sources link:

- Federal Highway Administration (FHWA) – Adaptive Signal Control Technologies
<https://highways.dot.gov/>
- Traffic Management Centers (TMCs)
<https://constanttech.com/what-is-a-traffic-management-center-and-how-does-it-work/>

- Road Network Operations & Intelligent Transport Systems (RNO & ITS)
<https://rno-its.piarc.org/en/systems-and-standards-its-architecture/what-its-architecture>
- United States Department of Transportation (U.S. DoT)
<https://www.itskrs.its.dot.gov/2018-b01259>
- Open Geospatial Consortium (OGC)
<https://www.ogc.org/who-we-are/>

3.1.2 Non-Functional Requirements (NFRs)

NFR-1 Performance: Requirement: ≤ 500 ms end-to-end latency; 95 % ≤ 500 ms under normal load (10 000 TPS). Being able to respond fast, reduce congested traffic, be able to alert accidents, and change its output accordingly.

NFR-2 Availability: 99.95 % availability without being affected by power outage by using solar panels, and using UPS, Have emergency maintaining team and regular maintaining team.

NFR-3 Scalability : Scale from 500 to 2 000 intersections while keeping ≤ 500 ms latency (verified in load testing). Being able to install more traffic lights while its output and response time is still fast and unaffected.

NFR-4 Security and Privacy: protect data when transferring using TLS 1.3 encryption, and using AES-256 storage to protect the stored information.

NFR-5 Disaster Recovery: able to recover its system not more than 15 minutes and not lost more than 5 minutes worth of information using RTO and RPO system

NFR-6 Mobile-App Performance: quick loading of application, information process fast without having to wait while the GPS or vehicle is moving continuously under the usage of 4G.

- **NFR-7 Interoperability:** Support National traffic light function of NTCIP 1202/1203, support communicating between server using Open AI and API
- Source: <https://www.ntcip.org/wp-content/uploads/2018/11/NTCIP1202v0219f.pdf>
- <https://idot.illinois.gov/content/dam/soi/en/web/idot/documents/transportation-system/research/success-stories/research-pays-off--safety-reports-and-statistics-page.pdf>

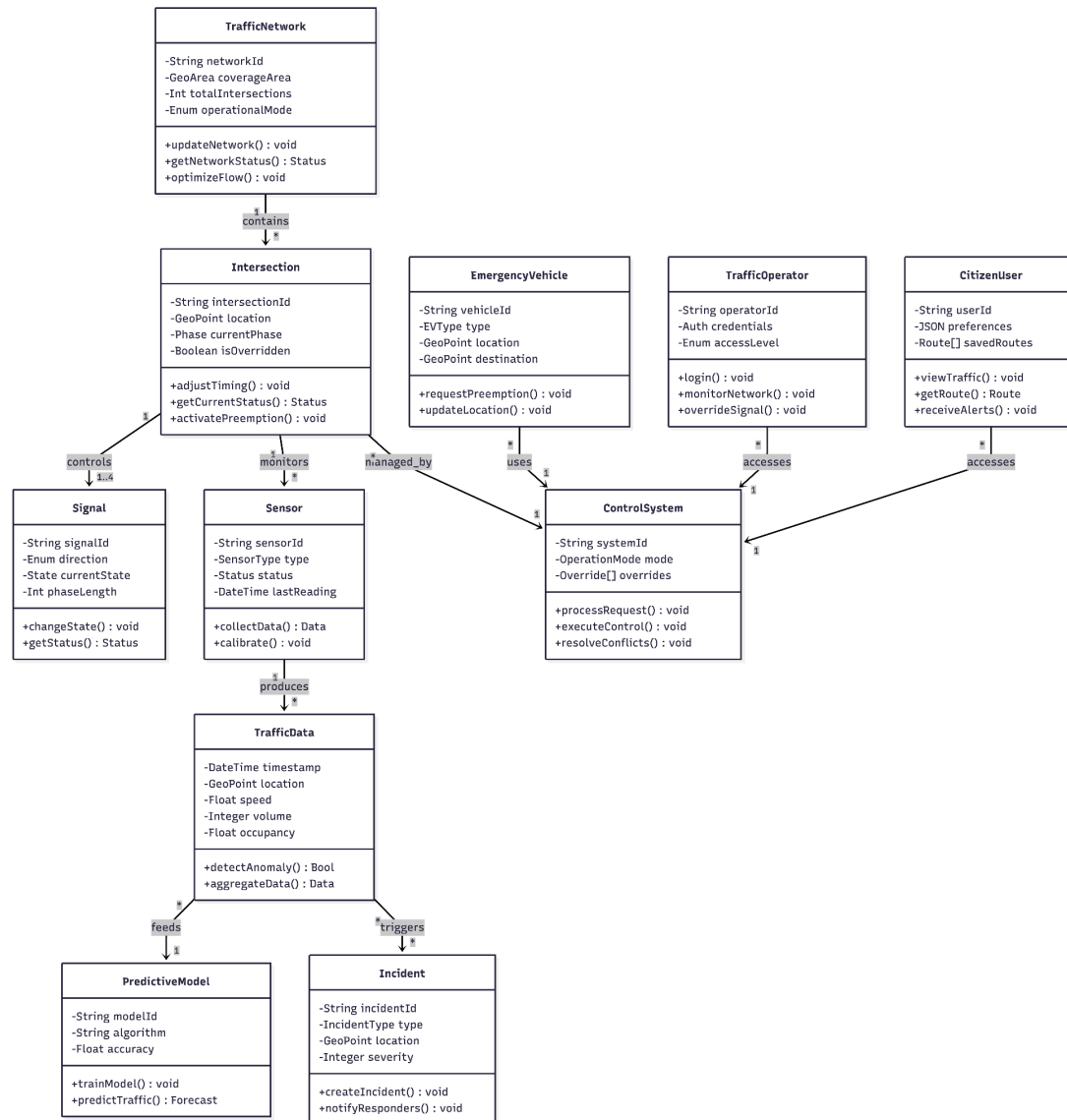
- <https://www.iotforall.com/ai-iot-traffic-design>
- <https://www.sciencedirect.com/science/article/pii/S2666691X20300105>

3.2 System Models and Glossary

3.2.1 UML Class Diagram

Purpose: Shows main system objects, their attributes, methods, and relationships

- **Traffic Network:** The top-level container managing all intersections
- **Intersection:** Controls signals and monitors sensors at each intersection
- **Signal & Sensor:** Physical devices that control lights and collect traffic data
- **Traffic Data:** Raw data collected from sensors
- **Predictive Model:** AI model that forecasts future traffic conditions
- **Control System:** Central brain coordinating all operations
- **Users:** Different types of users (operators, citizens, emergency vehicles)



3.2.2 UML Activity Diagram: Emergency Vehicle Preemption

Purpose: Shows the step-by-step process that happens when an emergency vehicle ambulance, fire truck, police car needs priority routing through traffic signals.

Start: Emergency vehicle activates preemption usually automatically when responding to a call

Validation: System checks if the vehicle is authorized prevents fake requests

Route Calculation: Finds the fastest route based on current traffic conditions

Intersection Processing: For each intersection along the route:

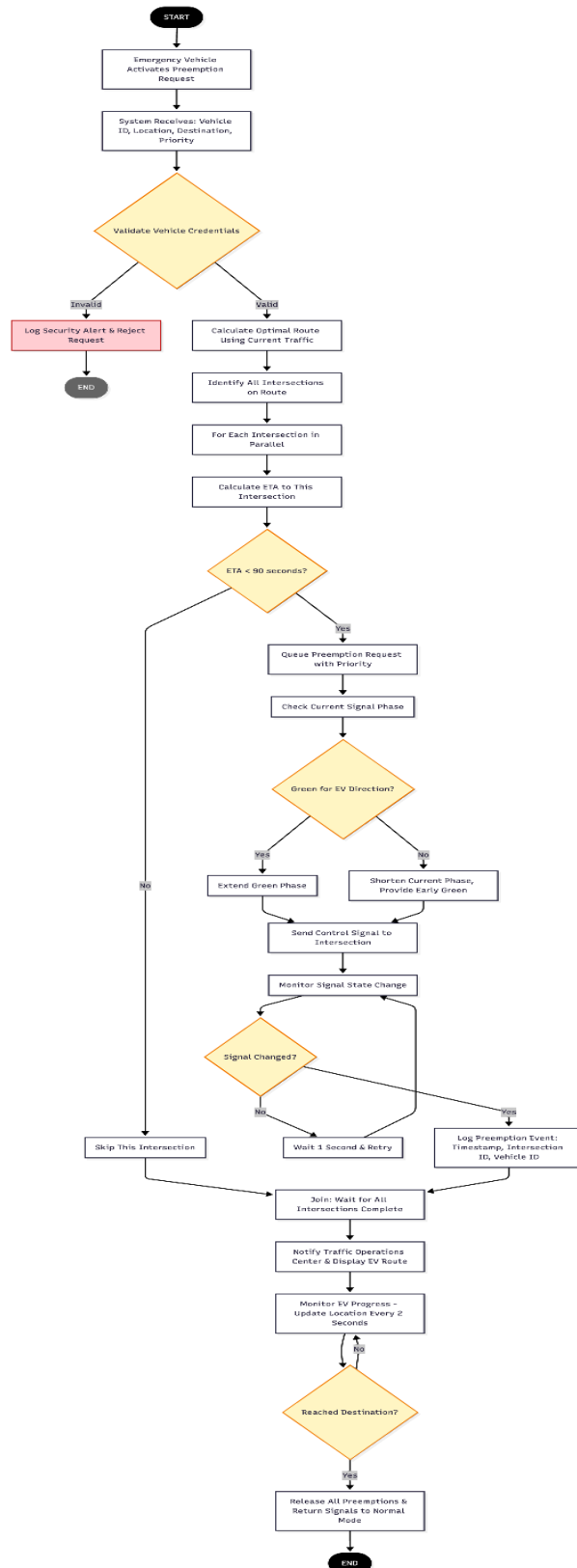
- Calculates when vehicle will arrive (ETA)
- If arriving in less than 90 seconds, prepares the signal
- Changes signal to green for the emergency vehicle direction
- Logs the event for analysis

Monitoring: Tracks vehicle progress continuously every 2 seconds.

Completion: When vehicle reaches destination, releases all preemptions and returns signals to normal operation.

Key Decision Points:

- **Valid credentials?** If no → Reject (security check)
- **ETA < 90 seconds?** If no → Skip intersection (too far away)
- **Green for EV?** Yes → Extend green time | No → Give early green
- **Signal changed?** If no → Retry (ensure command worked)



4.0 Phase 3: Strategy and Risk Management

4.1 Technical Feasibility and Architectural Vision

Overall Feasibility Assessment

Based on the technical analysis conducted in this study, the development of the Smart City Traffic Management System (SCTMS) is technically feasible. The required technologies already exist, are widely adopted, and have been successfully implemented in similar large-scale systems worldwide. While the project involves a high level of coordination and system integration, these challenges can be effectively managed through proper architectural design and planning.

Verdict: FEASIBLE

Several key factors support this conclusion:

- **Technology Maturity (High):** The system relies on proven technologies such as cloud computing platforms, Internet of Things (IoT) devices, edge computing, and artificial intelligence/machine learning frameworks. These technologies are stable, well-documented, and supported by major vendors, reducing technical uncertainty.
- **Technical Complexity (Medium–High):** Although the system is complex due to real-time processing, distributed components, and integration with legacy traffic infrastructure, this complexity is manageable with a well-structured architecture and experienced development teams.
- **Resources (Medium):** The estimated investment of approximately \$80–120 million and a Phase 1 implementation timeline of 18–24 months are reasonable for a smart city project of this scale. Adequate financial and human resources are assumed to be available.

- **Risk Level (Medium):** Identified technical risks, such as system integration and performance constraints, are considered manageable through proactive mitigation strategies outlined in this phase.

Overall, there are no major technological barriers that would prevent successful implementation of the proposed system.

Proposed Architecture: Event-Driven Microservices with Edge Computing

To meet the system's real-time performance, scalability, and reliability requirements, this project proposes an event-driven microservices architecture combined with edge computing. This hybrid edge-cloud approach balances fast local decision-making with centralized optimization and analytics.

The architecture consists of two main components:

- **Edge Computing at Intersections:** Critical, time-sensitive decisions—such as traffic signal adjustments and emergency vehicle preemption—are processed locally at each intersection. This ensures sub-second response times even when network connectivity to the cloud is limited or temporarily unavailable.
- **Cloud-Based Microservices:** Cloud services handle network-wide optimization, predictive analytics, long-term data storage, reporting, and integration with external systems. This enables coordinated traffic management across the entire city.

This architectural approach ensures high performance while maintaining flexibility and scalability.

Architectural Justification Based on Non-Functional Requirements

The proposed architecture directly supports the system's non-functional requirements as follows:

- **NFR-1 (Real-Time Performance < 500 ms):** Edge computing allows time-critical decisions to be processed locally, minimizing communication delays and ensuring fast responses.

- **NFR-2 (High Availability – 99.95% Uptime):** The microservices architecture isolates failures so that issues in one service do not affect the entire system. Additionally, edge components can continue operating autonomously if cloud services become unavailable.
- **NFR-3 (Scalability):** The system can scale horizontally by deploying additional microservice instances as traffic volume or the number of intersections increases, without requiring major architectural changes.
- **NFR-4 (Security and Privacy):** Sensitive data is anonymized at the edge before being transmitted to the cloud. Service isolation and secure communication protocols further enhance system security.
- **NFR-5 (Disaster Recovery):** Cloud services are deployed across multiple regions, enabling automatic failover and rapid recovery in the event of system failures.
- **NFR-6 (Mobile Performance):** Content Delivery Networks (CDNs) and data aggregation techniques reduce payload sizes, ensuring responsive mobile applications even on slower networks.
- **NFR-7 (Interoperability):** The use of standard APIs and industry protocols allows seamless integration with existing traffic controllers and third-party systems.

Architecture Layers

The overall system architecture is organized into multiple logical layers to ensure clarity, modularity, and maintainability:

- **Citizen / Public Access Layer**
This layer includes mobile applications (iOS and Android), a web portal, and a third-party navigation API gateway. It provides real-time traffic information and route recommendations to citizens and external services.
- **API Gateway and Load Balancer**
This layer manages authentication, rate limiting, request routing, and SSL termination. It acts as a secure entry point for all external requests.

- **Cloud Microservices Layer**

This layer consists of independent services responsible for user management, route calculation, incident management, predictive analytics, network optimization, emergency preemption, transit priority, environmental monitoring, and reporting. An event bus (e.g., Kafka) connects all services and enables asynchronous communication.

- **Edge Computing Layer (At Intersections)**

Each intersection hosts a local AI decision engine capable of making decisions in under 100 milliseconds. This layer interfaces directly with traffic signal controllers using NTCIP standards, aggregates sensor data, and includes an autonomous fallback mode that allows continued operation if cloud connectivity is lost.

- **Field Devices Layer**

This layer includes physical devices such as traffic sensors (loop detectors and radar), cameras, environmental sensors, and vehicle detection systems installed at intersections.

- **Data and Storage Layer**

Real-time sensor data is stored in a time-series database (e.g., InfluxDB), while relational databases (e.g., PostgreSQL) manage users, configurations, and system events. Long-term historical data is archived in a data lake for up to five years.

- **Operator / Administrator Interface**

A Traffic Management Center dashboard provides operators with real-time visibility, analytics, and manual control capabilities.

- **External Integrations Layer**

This layer connects the system with external services such as emergency dispatch systems (CAD), public transit tracking (AVL), weather services, and GIS/map providers.

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4.2 SRED Risk Analysis and Mitigation

Risk 1: Incomplete or Changing Stakeholder Requirements

One of the major risks in this project is that stakeholders may not fully express their needs at the beginning, or that they may request changes after development has already started. This can lead to delays, extra costs, and redesign work.

This risk is considered high likelihood and high impact, making it critical. To reduce this risk, the project will use an iterative requirements approach. Early mock-ups and simple prototypes will be developed during Weeks 4–6 and reviewed with stakeholders every two weeks. This allows misunderstandings to be discovered early, when changes are still easier and cheaper to make.

Requirements will also be prioritised using the MoSCoW method. “Must have” features will be formally approved and controlled, while lower-priority items can remain flexible. The system will be delivered in phases. A small pilot involving 10 intersections will be implemented first. Feedback from this phase will be used before expanding the system to 500 intersections.

Any requested change will go through a formal Change Control Board, which includes representatives from all stakeholder groups. Changes will only be approved after their cost and schedule impact has been reviewed.

In addition, a Requirements Traceability Matrix will be maintained to link each requirement to its source, design elements, and test cases. This will be checked monthly to ensure ongoing control.

Risk 2: Ambiguous or Unmeasurable Non-Functional Requirements

Another risk is that non-functional requirements such as performance, reliability, or availability may be written in vague terms (for example, “the system should be fast”), which makes them difficult to test and enforce.

This risk has a medium-high likelihood and medium impact, giving it a high risk level. To manage this, a standard NFR template will be used. Every non-functional requirement must include a clear metric, a target value, and a method of testing. Workshops will be organised with both technical staff and business stakeholders to agree on realistic and measurable values, based on how the system will actually be used.

The project team will also carry out benchmarking, including measuring current systems and reviewing recognised standards such as IEEE and NIST. A small prototype will be built early in the project to check whether these targets are achievable before full-scale development begins.

Finally, key NFRs will be included in vendor contracts, with defined acceptance tests and performance criteria.

Risk 3: Stakeholder Availability and Engagement

Some stakeholders may not always be available due to their daily operational responsibilities. This could slow down decision-making and weaken requirements quality.

This risk is rated as medium likelihood and medium-high impact, making it high. To address this, executive sponsorship will be secured to ensure that stakeholder participation is formally supported at management level. Meetings will be designed to be time-efficient, with short sessions, clear agendas, and prepared materials. When necessary, feedback will also be collected asynchronously. Each stakeholder group will nominate representatives (“champions”) who are authorised to provide input and make routine decisions.

A focused requirements sprint will be run over four weeks to maintain momentum and ensure consistent engagement. Participation will be encouraged by making involvement valuable, such as providing early access to the pilot system and opportunities to review real deployments.

Risk 4: Integration with Legacy Traffic Control Systems

Many existing traffic systems are old, use proprietary technologies, or lack proper documentation. This makes integration technically difficult and risky.

This risk is considered high likelihood and high impact, and therefore critical. The project will begin with a full audit of all intersections to document hardware, software, and communication protocols. An early proof-of-concept will then be developed using a small number of representative intersections. This will help identify serious technical issues before final architecture decisions are made.

The system will use a layered integration approach, where the core platform communicates through adapters rather than directly with controllers. A contingency budget will be set aside for replacing incompatible hardware over time. The project will also work closely with equipment vendors, including negotiating technical support and specialist assistance where required.

Risk 5: Data Quality and Sensor Reliability

Traffic data may be unreliable due to sensor faults, environmental conditions, or physical damage. Poor data quality could reduce system accuracy and trust. This risk has a medium likelihood and medium-high impact, giving it a high rating. To reduce this risk, multiple sensor types will be deployed at key intersections so that readings can be compared and validated. The system will include automatic data quality checks to detect unusual values, failed sensors, and inconsistencies. All system outputs will include a confidence indicator, and conservative control strategies will be used when confidence is low.

A proactive maintenance schedule will be established, including regular inspections and calibration. In addition, users will be able to report issues through a mobile interface, allowing crowdsourced validation of system data.

Risk Summary Matrix

Risk	Likelihood	Impact	Priority	Mitigation Cost
Changing Requirements	HIGH	HIGH	CRITICAL	\$150K

Unmeasurable NFRs	MED-HIGH	MEDIUM	HIGH	\$50K
Stakeholder Availability	MEDIUM	MED-HIGH	HIGH	\$200K
Legacy Integration	HIGH	HIGH	CRITICAL	\$3M+
Data Quality	MEDIUM	MED-HIGH	HIGH	\$500K

Total Mitigation Budget: \$3.9M (8% of Phase 1 budget)

5.0 Conclusion and Next Steps

Summary

This feasibility study confirms that a Smart City Traffic Management System is urgently needed and technically achievable.

Key Findings:

- Clear Value: \$30-40M annual savings, 15-25% travel time reduction, 20-30% faster emergency response
- Well-Defined Requirements: 15 FRs and 7 measurable NFRs from 7 stakeholder groups
- Robust Architecture: Event-driven microservices with edge computing meets all critical NFRs
- Manageable Risks: Two critical risks (requirements volatility, legacy integration) with comprehensive mitigation plans

Recommendations

Immediate Actions (Next 30 Days):

- Secure executive sponsorship and budget authorization
- Form project governance (steering committee, change control board)
- Launch legacy system audit
- Issue RFI to vendors

Phase 1 Implementation (Months 1-24):

- Months 1-3: Detailed design, vendor selection
- Months 4-9: 10-intersection pilot deployment
- Months 10-18: Expand to 500 intersections Months
- 19-24: Stabilization and evaluation

Success Criteria:

- Achieve all NFRs (99.95% uptime, <500ms latency)
- 15% travel time reduction
- 50,000+ mobile app downloads
- Stay within \$50M budget (+/- 10%)

We strongly recommend proceeding immediately to Phase 1 Implementation.

Appendix A: Glossary of Terms

Term	Definition
Adaptive Signal Control	Traffic signal timing that automatically adjusts based on realtime conditions
Arterial Corridor	Major road carrying large traffic volumes with multiple signals
AVL (Automatic Vehicle Location)	GPS-based system tracking real-time vehicle position
CAD (Computer-Aided Dispatch)	Software for emergency services to manage incident reports
Congestion	Traffic condition where demand exceeds capacity, causing delays
Connected Vehicle (V2X)	Vehicle with communication technology to exchange data
Cycle Length	Total time for signal to complete all phases (typically 60-120 seconds)

Edge Computing	Data processing near source rather than centralized cloud
Emergency Vehicle Preemption	Override normal signals to provide priority green lights
Geofencing	Virtual boundary triggering actions when device enters/exits
Green Wave	Coordinated signals allowing consecutive green lights at specific speed
Incident	Unplanned event disrupting traffic (accident, breakdown, debris)
Intersection	Junction where two or more roadways meet
Latency	Time delay between data collection and control action
Level of Service (LOS)	Traffic quality measure graded A (free flow) to F (gridlock)
Microservices	Software built as collection of small independent services
NTCIP	National Transportation Communications for ITS Protocol
Occupancy	Percentage of time detection zone occupied by vehicles
Phase	Period when specific traffic movement is allowed
Predictive Analytics	Using historical data and ML to forecast future conditions
Queue Length	Number of vehicles waiting at red light
Real-Time System	Computing system with strict time constraints (milliseconds to seconds)
Signal Timing Plan	Configuration of phase durations and transitions
Throughput	Volume of vehicles passing through per time unit

Traffic Management Center	Facility where operators monitor and control network
Transit Signal Priority	Extending greens or shortening reds for transit vehicles
Vehicle Detection	Sensors identifying vehicle presence, speed, characteristics
V/C Ratio	Volume-to-capacity ratio; above 1.0 indicates oversaturation
WCAG	Web Content Accessibility Guidelines for people with disabilities

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END OF REPORT

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